

Homework 5 Vulnerability Reports and MCP Tools

Name: Ayushi Shringi

SID4: 0036

PORT_BASE: 8036

PREFIX: s0036

SEED: 36

VERIFY_SEED: 260036

DOMAIN_ID: 4

Hardware: MacBook Pro, Apple M5 Pro, 24 GB

Model: qwen3:4b

GitHub Repository: <https://github.com/ayushishringi/data260-0036/tree/hw5>

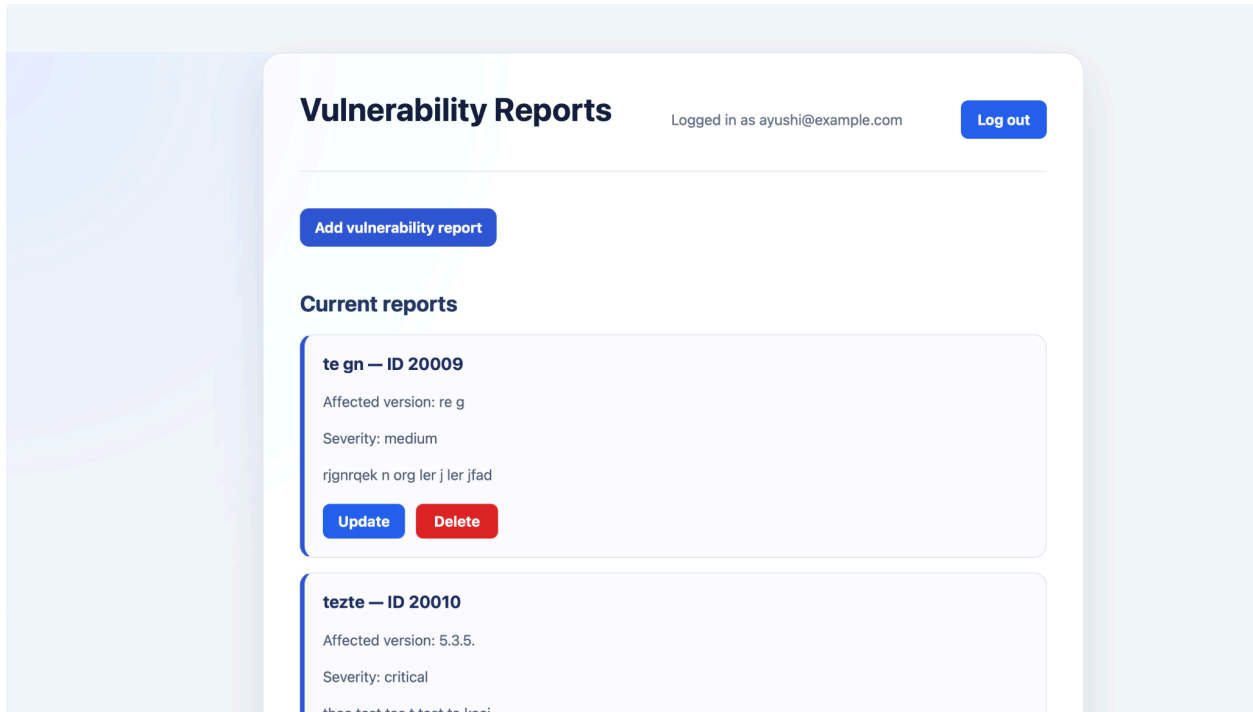
Tag: hw5

Commit: 02672d93befa5c559f93698fd8d3a1881461d7f1

The GitHub repository link was verified, and both collaborators, have access.

Home Screen

```
frontend > src > JS store.js > ...
1  import { configureStore, createAsyncThunk, createSlice } from "@reduxjs/toolkit";
2  import axios from "axios";
3  import { API_BASE } from "../api";
4  const API = API_BASE;
5  const request = (config) => axios({ ...config, baseURL: API, withCredentials: true });
6  const failure = (error) => error.response?.data?.detail || error.message;
7  export const fetchReports = createAsyncThunk("reports/fetch", async (_, {
    rejectWithValue }) => { try { return (await request({ url: "/api/reports" })).data; }
    catch (e) { return rejectWithValue(failure(e)); } });
8  export const createReport = createAsyncThunk("reports/create", async (payload, {
    rejectWithValue }) => { try { return (await request({ method: "post", url: "/api/
    reports", data: payload })).data; } catch (e) { return rejectWithValue(failure(e)); } });
9  export const updateReport = createAsyncThunk("reports/update", async ({ id, payload }, {
    rejectWithValue }) => { try { return (await request({ method: "put", url: `/api/reports/$
    {id}`, data: payload })).data; } catch (e) { return rejectWithValue(failure(e)); } });
10 export const deleteReport = createAsyncThunk("reports/delete", async (id, {
    rejectWithValue }) => { try { await request({ method: "delete", url: `/api/reports/${id}
    ` }); return id; } catch (e) { return rejectWithValue(failure(e)); } });
11
12 const slice = createSlice({ name: "reports", initialState: { items: [], loading: false,
    error: "" }, reducers: {}, extraReducers: (builder) => builder
13   .addCase(fetchReports.pending, (s) => { s.loading = true; s.error = ""; })
14   .addCase(fetchReports.fulfilled, (s, a) => { s.loading = false; s.items = a.payload; })
15   .addCase(fetchReports.rejected, (s, a) => { s.loading = false; s.error = a.payload ||
    "Request failed"; })
16   .addCase(createReport.fulfilled, (s, a) => { s.items.push(a.payload); })
17   .addCase(updateReport.fulfilled, (s, a) => { const i = s.items.findIndex((r) => r.id
    === a.payload.id); if (i >= 0) s.items[i] = a.payload; })
18   .addCase(deleteReport.fulfilled, (s, a) => { s.items = s.items.filter((r) => r.id !==
    a.payload); })
19   .addMatcher((a) => a.type.startsWith("reports/") && a.type.endsWith("/rejected"), (s,
    a) => { s.error = a.payload || "Request failed"; } ));
20 export const store = configureStore({ reducer: { reports: slice.reducer } });
21
```



Redux Create Report

```
catch (e) { return rejectWithValue(failure(e)); } }},  
export const createReport = createAsyncThunk("reports/create", async (payload, {  
  rejectWithValue }) => { try { return (await request({ method: "post", url: "/api/  
  reports", data: payload })).data; } catch (e) { return rejectWithValue(failure(e)); } }));
```

localhost:5173/create

Document - Google Docs

fr

Add Vulnerability Report

← Back to reports

Package name

ttest_12

Affected version

1.3.41

Submitter email

ayushi@example.com

Description

this is a test report. jsut of r the we saw. red

Severity

High

☒

I agree to the terms.

Submit report

Update and Delete Report

```
export const updateReport = createAsyncThunk("reports/update", async ({ id, payload }, {
  rejectWithValue }) => { try { return (await request({ method: "put", url: `/api/reports/${
    id}`, data: payload })).data; } catch (e) { return rejectWithValue(failure(e)); } });
export const deleteReport = createAsyncThunk("reports/delete", async (id, {
  rejectWithValue }) => { try { await request({ method: "delete", url: `/api/reports/${id}
  ` }); return id; } catch (e) { return rejectWithValue(failure(e)); } });
```

localhost:5173/update/20009

Untitled document - Google Docs

frontend

Update Vulnerability Report

[← Back to reports](#)

Package name

te gn

Affected version

re g

Submitter email

ayushi@example.com

Description

rjgnrqek n org ler j ler jfad

Severity

Medium

☒

I agree to the terms.

Current reports

te gn — ID 20009

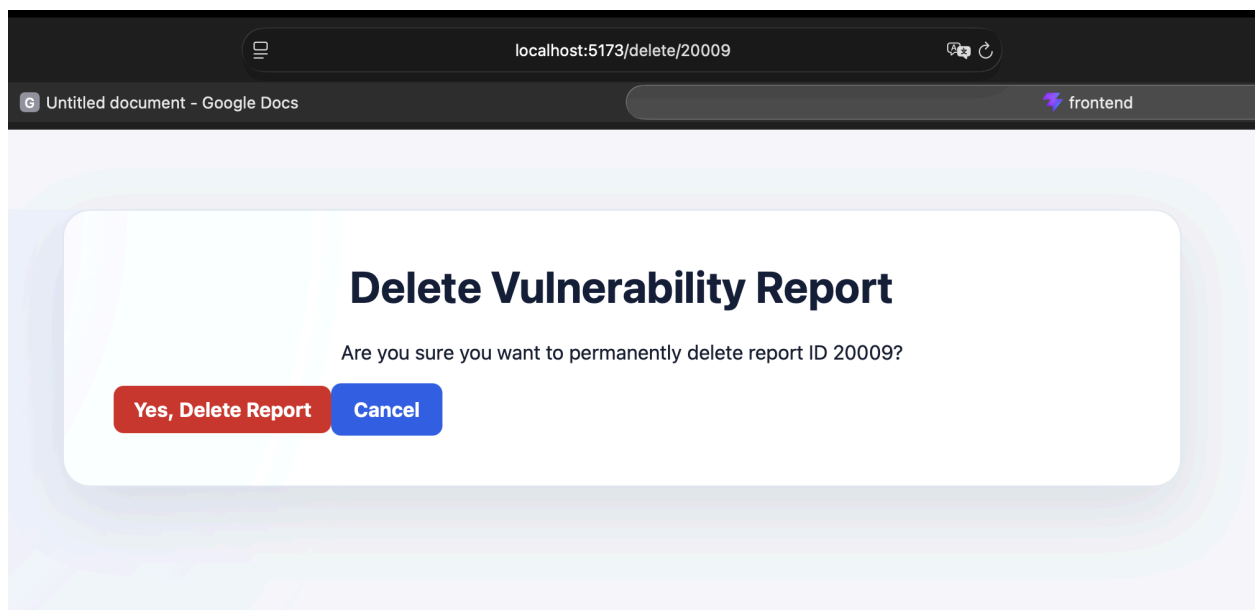
Affected version: re g

Severity: high

rjgnrqek n org ler j ler jfad

Update

Delete



Report CRUD Route Implementation

```

52 @router.get("")
53 def list_reports(
54     db: Session = Depends(get_db),
55     current_user: User = Depends(get_current_user),
56 ) -> List[dict]:
57     reports = db.scalars(
58         select(VulnerabilityReport)
59         .options(
60             selectinload(VulnerabilityReport.advisories),
61             selectinload(VulnerabilityReport.advisory),
62         )
63         .order_by(VulnerabilityReport.id)
64     ).all()
65
66
67     return [report_to_dict(report) for report in reports]
68
69
70 @router.get("/{report_id}")
71 def get_report(
72     report_id: int,
73     db: Session = Depends(get_db),
74     current_user: User = Depends(get_current_user),
75 ) -> dict:
76     report = db.scalar(
77         select(VulnerabilityReport)
78         .options(
79             selectinload(VulnerabilityReport.advisories),
80             selectinload(VulnerabilityReport.advisory),
81         )
82         .where(VulnerabilityReport.id == report_id)
83     )
84
85     if report is None:
86         raise HTTPException(
87             status_code=404,
88             detail="Report not found",
89         )
90
91     return report_to_dict(report)
92
93
94 @router.post("", status_code=201)
95 def create_report(
96     payload: ReportInput,

```

```

95     def create_report(
96     ) -> dict:
97         report = VulnerabilityReport(
98             package_name=payload.packageName,
99             affected_version=payload.affectedVersion,
100             submitter_email=str(payload.submitterEmail),
101             description=payload.description,
102             severity=payload.severity,
103             agreed_to_terms=payload.agreedToTerms,
104             submission_date=datetime.now(
105                 timezone.utc
106             ).replace(tzinfo=None),
107             advisory_id=payload.advisoryId,
108             available_count=payload.availableCount,
109         )
110
111         db.add(report)
112         db.commit()
113         db.refresh(report)
114
115         return report_to_dict(report)
116
117
118 @router.put("/{report_id}")
119 def update_report(
120     report_id: int,
121     payload: ReportInput,
122     db: Session = Depends(get_db),
123     current_user: User = Depends(get_current_user),
124 ) -> dict:
125     report = db.get(VulnerabilityReport, report_id)
126
127     if report is None:
128         raise HTTPException(
129             status_code=404,
130             detail="Report not found",
131         )
132
133     report.package_name = payload.packageName
134     report.affected_version = payload.affectedVersion
135     report.submitter_email = str(payload.submitterEmail)
136     report.description = payload.description
137     report.severity = payload.severity
138     report.agreed_to_terms = payload.agreedToTerms
139     report.advisory_id = payload.advisoryId

```

Advisory CRUD Route Implementation

```
backend > report_api.py > report_to_dict
168 @router.get("/advisories")
169 def list_advisories(
170     offset: int = 0,
171     limit: int = 20,
172     db: Session = Depends(get_db),
173     current_user: User = Depends(get_current_user),
174 ) -> list[dict]:
175     limit = min(max(limit, 1), 100)
176     advisories = db.scalars(
177         select(Advisory).order_by(Advisory.id).offset(max(offset, 0)).limit(limit)
178     ).all()
179     return [advisory_to_dict(item) for item in advisories]
180
181
182 @router.get("/advisories/{advisory_id}")
183 def get_advisory(
184     advisory_id: int,
185     db: Session = Depends(get_db),
186     current_user: User = Depends(get_current_user),
187 ) -> dict:
188     advisory = db.get(Advisory, advisory_id)
189     if advisory is None:
190         raise HTTPException(status_code=404, detail="Advisory not found")
191     return advisory_to_dict(advisory)
192
193
194 @router.post("/advisories", status_code=201)
195 def create_advisory(
196     payload: AdvisoryInput,
197     db: Session = Depends(get_db),
198     current_user: User = Depends(get_current_user),
199 ) -> dict:
200     if db.scalar(select(Advisory).where(Advisory.advisory_code == payload.advisoryCode)):
201         raise HTTPException(status_code=409, detail="Advisory code already exists")
202     advisory = Advisory(
203         name=payload.name.strip(), publisher=payload.publisher.strip(),
204         advisory_code=payload.advisoryCode.strip(),
205     )
206     db.add(advisory)
207     db.commit()
208     db.refresh(advisory)
209     return advisory_to_dict(advisory)
210
211
212 @router.put("/advisories/{advisory_id}")
213 def update_advisory(
```

```

backend > report_api.py > report_to_dict
213 def update_advisory(
219     raise HTTPException(status_code=404, detail="Advisory not found")
220     duplicate = db.scalar(select(Advisory).where(
221         Advisory.advisory_code == payload.advisoryCode, Advisory.id != advisory_id
222     ))
223     if duplicate:
224         raise HTTPException(status_code=409, detail="Advisory code already exists")
225     advisory.name, advisory.publisher, advisory.advisory_code = (
226         payload.name.strip(), payload.publisher.strip(), payload.advisoryCode.strip()
227     )
228     db.commit()
229     db.refresh(advisory)
230     return advisory_to_dict(advisory)
231
232
233 @router.delete("/advisories/{advisory_id}", status_code=204)
234 def delete_advisory(
235     advisory_id: int, db: Session = Depends(get_db),
236     current_user: User = Depends(get_current_user),
237 ) -> None:
238     advisory = db.get(Advisory, advisory_id)
239     if advisory is None:
240         raise HTTPException(status_code=404, detail="Advisory not found")
241     if db.scalar(select(VulnerabilityReport.id).where(VulnerabilityReport.advisory_id == advisory_id)):
242         raise HTTPException(status_code=409, detail="Cannot delete advisory with linked reports")
243     db.delete(advisory)
244     db.commit()
245
246
247 @router.get("/advisories/{advisory_id}/reports")
248 def reports_for_advisory(
249     advisory_id: int, db: Session = Depends(get_db),
250     current_user: User = Depends(get_current_user),
251 ) -> list[dict]:
252     if db.get(Advisory, advisory_id) is None:
253         raise HTTPException(status_code=404, detail="Advisory not found")
254     reports = db.scalars(select(VulnerabilityReport).where(
255         VulnerabilityReport.advisory_id == advisory_id
256     ).order_by(VulnerabilityReport.id)).all()
257     return [report_to_dict(report) for report in reports]
258
259
260 @router.delete("/{report_id}", status_code=204)
261 def delete_report(
262     report_id: int,
263     db: Session = Depends(get_db),
264     current_user: User = Depends(get_current_user),
265 ) -> None:
266     report = db.get(VulnerabilityReport, report_id)
267
268     if report is None:
269         raise HTTPException(
270             status_code=404,
271             detail="Report not found",
272         )
273

```

Advisory–Report Relationship Query

```

239     if advisory is None:
240         raise HTTPException(status_code=404, detail="Advisory not found")
241     if db.scalar(select(VulnerabilityReport.id).where(VulnerabilityReport.advisory_id == a
242         raise HTTPException(status_code=409, detail="Cannot delete advisory with linked re
243     db.delete(advisory)
244     db.commit()
245
246
247 @router.get("/advisories/{advisory_id}/reports")
248 def reports_for_advisory(
249     advisory_id: int, db: Session = Depends(get_db),
250     current_user: User = Depends(get_current_user),
251 ) -> list[dict]:
252     if db.get(Advisory, advisory_id) is None:
253         raise HTTPException(status_code=404, detail="Advisory not found")
254     reports = db.scalars(select(VulnerabilityReport).where(
255         VulnerabilityReport.advisory_id == advisory_id
256     ).order_by(VulnerabilityReport.id)).all()
257     return [report_to_dict(report) for report in reports]
258
259
260 @router.delete("/{report_id}", status_code=204)
261 def delete_report(
262     report_id: int,
263     db: Session = Depends(get_db),
264     current_user: User = Depends(get_current_user),
265 ) -> None:
266     report = db.get(VulnerabilityReport, report_id)
267

```

Get Reports API

HTTP <http://localhost:8036/api/reports> Save Share

GET <http://localhost:8036/api/reports?offset=0&limit=10> Send

Docs Params Authorization Headers 7 Body Scripts Settings Cookies

Body Cookies 1 Headers 5 Test Results 200 OK • 154 ms • 2.18 MB

{ JSON Preview Visualize

```
1 [
2   {
3     "id": 15008,
4     "reportCode": "VULN-0036-00015008",
5     "packageName": "hw4-n1-package-1",
6     "affectedVersion": "< 1.0.0",
7     "submitterEmail": "benchmark@example.com",
8     "description": "Benchmark vulnerability description for measuring SQL query performance.",
9     "severity": "medium",
10    "agreedToTerms": true,
11    "submissionDate": "2026-09-27T05:01:44",
12    "createdAt": "2026-10-04T16:18:55",
13    "updatedAt": "2026-10-05T11:59:34",
14    "advisoryId": null,
15    "availableCount": 1,
16    "advisory": null,
17    "related": [
18      {
19        "id": 602,
20        "text": "Benchmark advisory text for report 1."
21      }
22    ]
23  },
24  {
25    "id": 15009,
26    "reportCode": "VULN-0036-00015009",
27    "packageName": "hw4-n1-package-2",
28    "affectedVersion": "< 1.0.0",
29    "submitterEmail": "benchmark@example.com",
30    "description": "Benchmark vulnerability description for measuring SQL query performance.",
31    "severity": "medium",
32    "agreedToTerms": true,
33    "submissionDate": "2026-09-27T05:01:44",
34    "createdAt": "2026-10-04T16:18:55",
35    "updatedAt": "2026-10-05T11:59:34",
36    "advisoryId": null,
37    "availableCount": 1
```

Create Advisory API

Post data

POST http://localhost:8036/api/reports/advisories

Docs Params Authorization Headers 9 Body Scripts Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON Schema Beautify

```
1 {
2   "name": "Report Security Advisory",
3   "publisher": "Test Publisher",
4   "advisoryCode": "REPORT-0036"
5 }
```

Body Cookies 1 Headers 4 Test Results 1/1 201 Created • 23 ms • 301 B Save Response

{ } JSON Preview Visualize

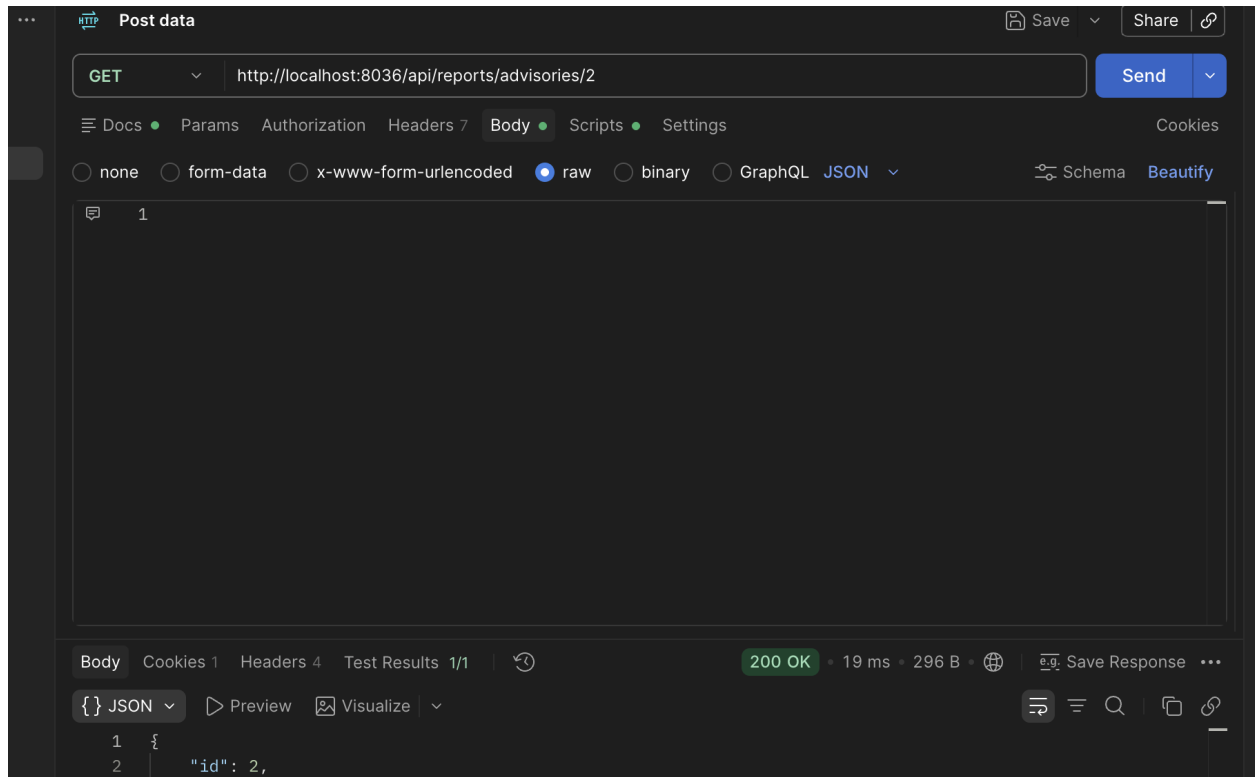
```
1 {
2   "id": 2,
3   "name": "Report Security Advisory",
4   "publisher": "Test Publisher",
5   "advisoryCode": "REPORT-0036",
6   "createdAt": "2026-10-04T09:12:27",
7   "updatedAt": "2026-10-04T09:12:27"
8 }
```

List Advisories API

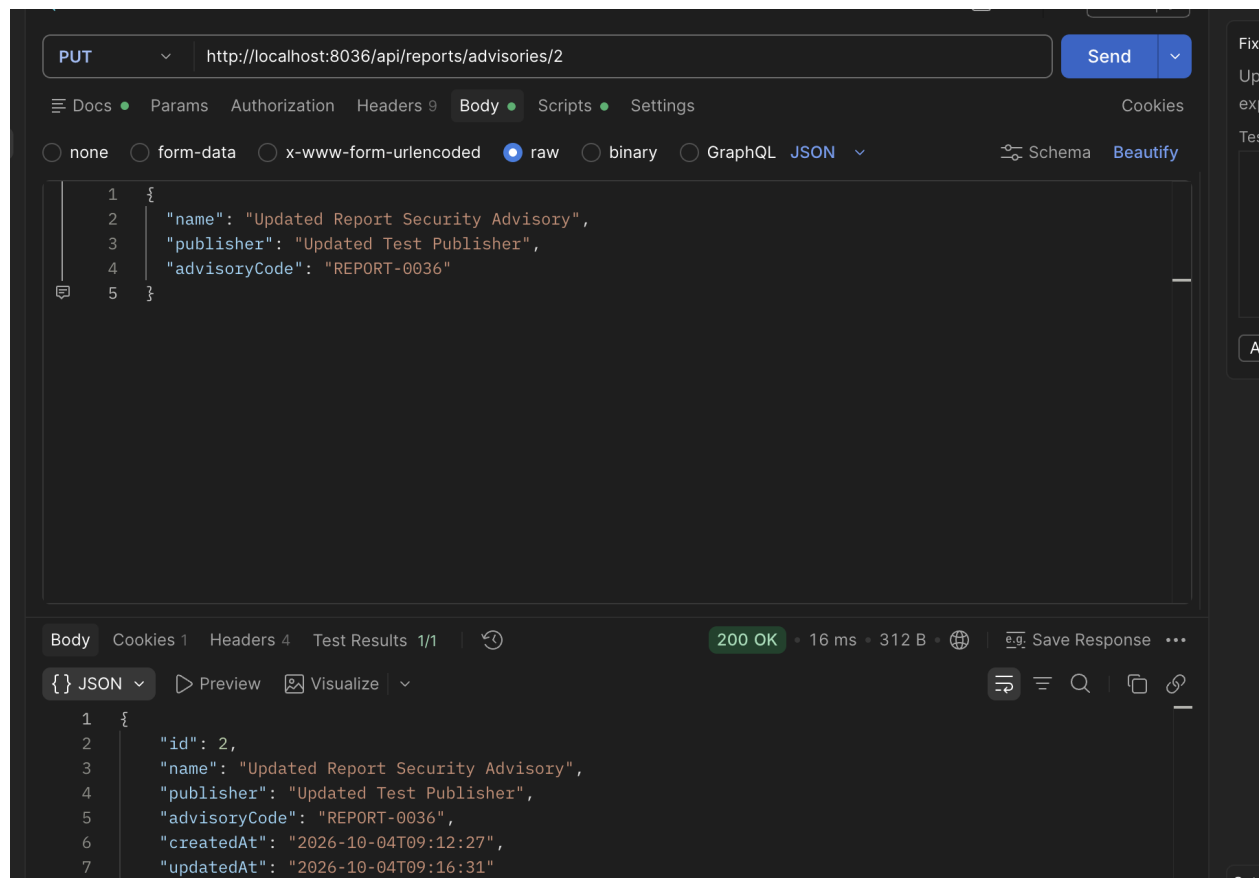
The screenshot shows a web browser's developer tools interface. At the top, the 'Post data' tab is active, displaying a GET request to the URL `http://localhost:8036/api/reports/advisories?offset=0&limit=10`. The 'Body' tab is selected, showing the request body as 'raw'. Below the request, the 'Response' tab is active, displaying a 200 OK status with a response time of 3 ms and a size of 298 B. The response body is a JSON array containing one object representing a security advisory.

```
{
  "id": 2,
  "name": "Report Security Advisory",
  "publisher": "Test Publisher",
  "advisoryCode": "REPORT-0036",
  "createdAt": "2026-10-04T09:12:27",
  "updatedAt": "2026-10-04T09:12:27"
}
```

Get Advisory API



Update Advisory API



Create API

POST http://localhost:8036/api/reports Send

Docs Params Authorization Headers 9 Body Scripts Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON Schema Beautify

```
1 {
2   "packageName": "api-report-test",
3   "affectedVersion": "1.0.0",
4   "submitterEmail": "ayushi@example.com",
5   "description": "This report demonstrates the FastAPI vulnerability report create endpoint.",
6   "severity": "high",
7   "agreedToTerms": true,
8   "advisoryId": 2,
9   "availableCount": 2
10 }
```

Body Cookies 1 Headers 4 Test Results 1/1 201 Created • 20 ms • 585 B Save Response

{ } JSON Preview Visualize

```
2   "id": 20013,
3   "packageName": "api-report-test",
4   "affectedVersion": "1.0.0",
5   "submitterEmail": "ayushi@example.com",
6   "description": "This report demonstrates the FastAPI vulnerability report create endpoint.",
7   "severity": "high",
8   "agreedToTerms": true,
9   "submissionDate": "2026-10-04T09:17:43",
10  "advisoryId": 2,
11  "availableCount": 2,
```

List Report API

GET http://localhost:8036/api/reports Send

Docs Params Authorization Headers 7 Body Scripts Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON Schema Beautify

1

Body Cookies 1 Headers 4 Test Results 1/1 200 OK • 133 ms • 1.69 MB Save Response

{ } JSON Preview Visualize

```
1 [
2   {
3     "id": 15007,
4     "packageName": "hw4-n1-package-0",
5     "affectedVersion": "< 1.0.0",
6     "submitterEmail": "benchmark@example.com",
7     "description": "Benchmark vulnerability description for measuring SQL query performance.",
8     "severity": "medium",
9     "agreedToTerms": true,
10    "submissionDate": "2026-09-27T05:01:44",
11    "advisoryId": null,
12    "availableCount": 1,
13    "advisory": null,
14    "related": [
15      {
16        "id": 601,
17        "text": "Benchmark advisory text for report 0."
18      }
19    ]
20  },
21  {
22    "id": 15008,
23    "packageName": "hw4-n1-package-1",
24    "affectedVersion": "< 1.0.0",
25    "submitterEmail": "benchmark@example.com",
```

Get Report API

GET http://localhost:8036/api/reports/15007 Send

Docs Params Authorization Headers 7 Body Scripts Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON Schema Beautify

1

Body Cookies 1 Headers 4 Test Results 1/1 200 OK • 15 ms • 534 B • Save Response ...

{ } JSON Preview Visualize

```
1 {
2   "id": 15007,
3   "packageName": "hw4-n1-package-0",
4   "affectedVersion": "< 1.0.0",
5   "submitterEmail": "benchmark@example.com",
6   "description": "Benchmark vulnerability description for measuring SQL query performance.",
7   "severity": "medium",
8   "agreedToTerms": true,
9   "submissionDate": "2026-09-27T05:01:44",
10  "advisoryId": null,
11  "availableCount": 1,
12  "advisory": null,
13  "related": [
14    {
15      "id": 601,
16      "text": "Benchmark advisory text for report 0."
17    }
18  ]
19 }
```

Update Report API

PUT http://localhost:8036/api/reports/15007 Send

Docs Params Authorization Headers 9 Body Scripts Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON Schema Beautify

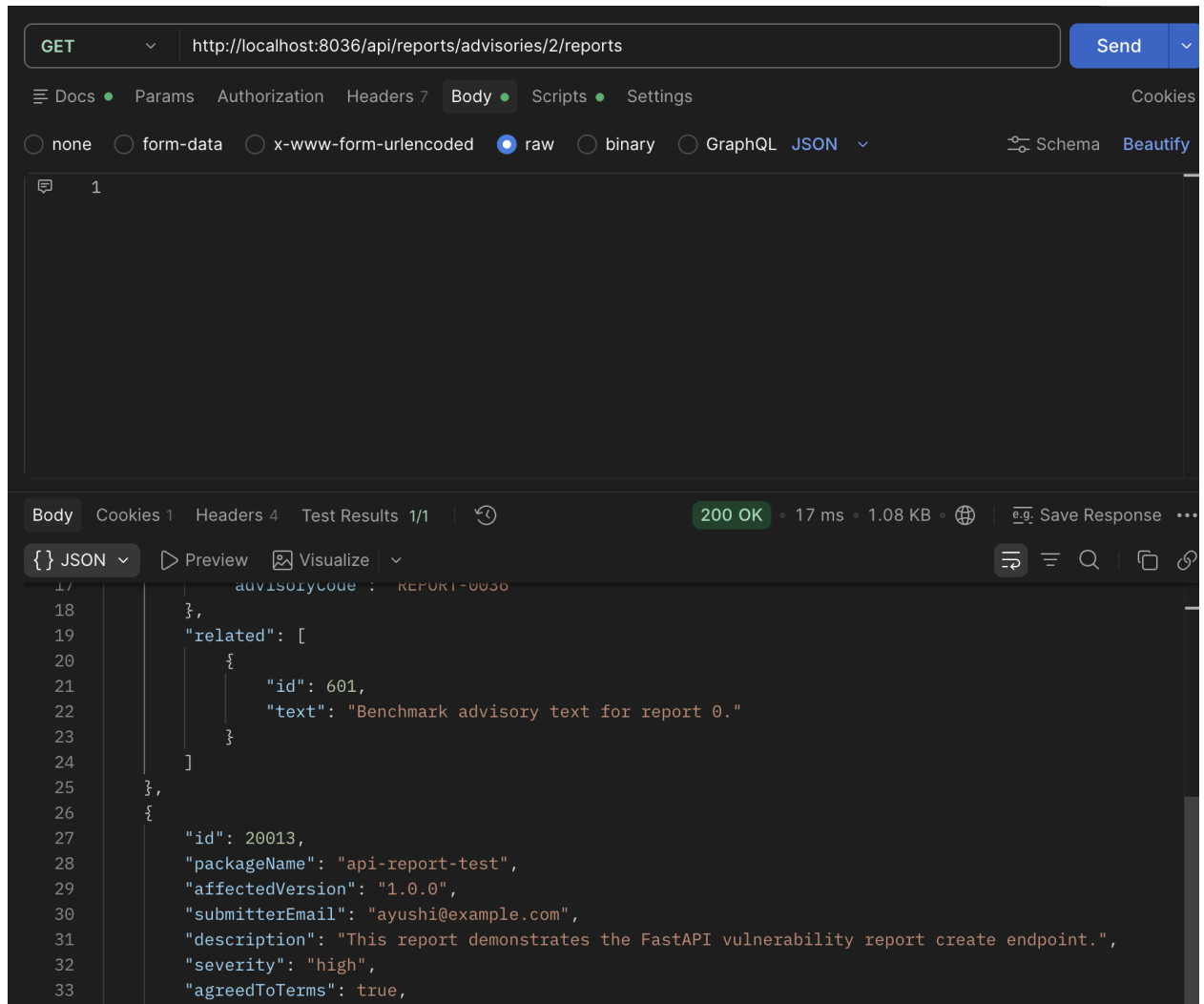
```
1 {
2   "packageName": "api-report-test-updated",
3   "affectedVersion": "2.0.0",
4   "submitterEmail": "ayushi@example.com",
5   "description": "This report demonstrates updating a vulnerability through the FastAPI endpoint.",
6   "severity": "critical",
7   "agreedToTerms": true,
8   "advisoryId": 2,
9   "availableCount": 5
10 }
```

Body Cookies 1 Headers 4 Test Results 1/1 200 OK • 23 ms • 654 B • Save Response

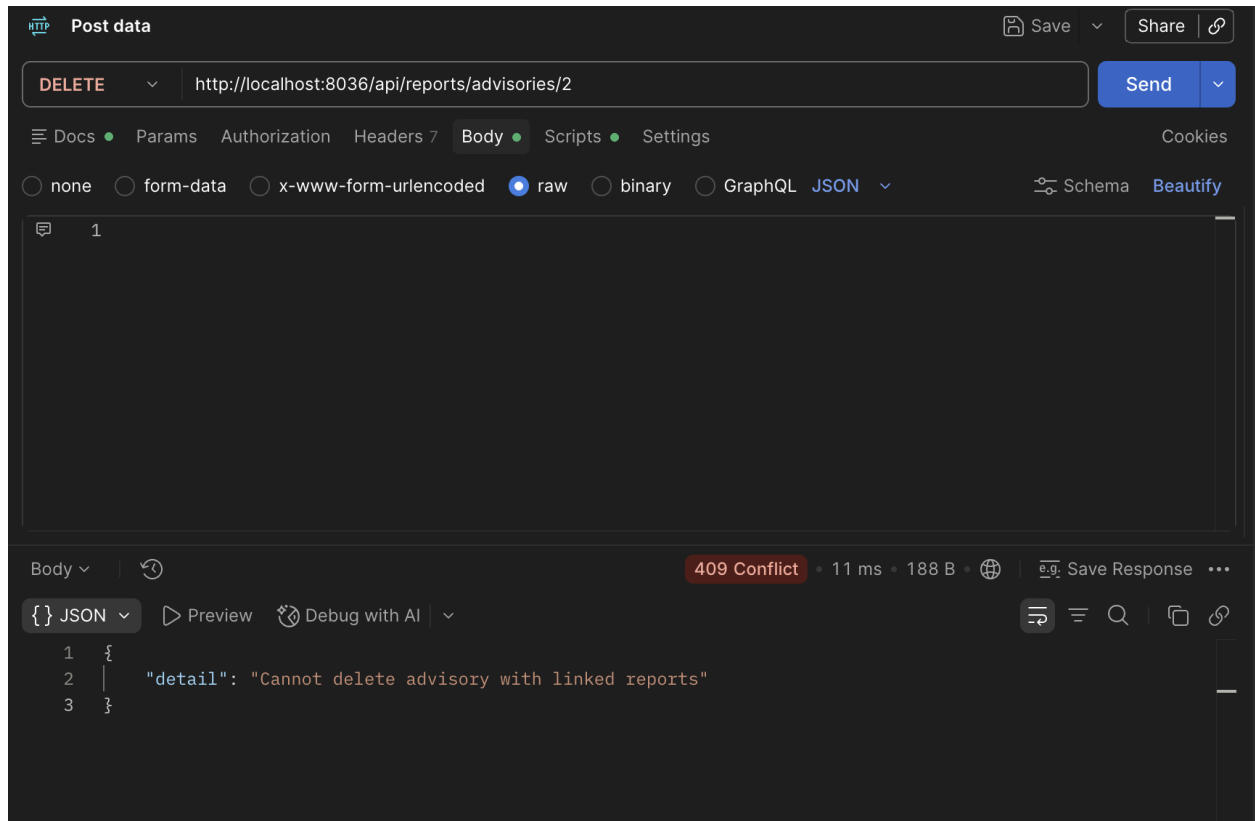
{ } JSON Preview Visualize

```
1 {
2   "id": 15007,
3   "packageName": "api-report-test-updated",
4   "affectedVersion": "2.0.0",
5   "submitterEmail": "ayushi@example.com",
6   "description": "This report demonstrates updating a vulnerability through the FastAPI endpoint.",
7   "severity": "critical",
8   "agreedToTerms": true,
9   "submissionDate": "2026-09-27T05:01:44",
10  "advisoryId": 2,
11  "availableCount": 5,
12  "advisory": {
13    "id": 2,
14    "name": "Updated Report Security Advisory",
15    "publisher": "Updated Test Publisher",
16    "advisoryCode": "REPORT-0036"
17  },
18  "related": [
19    {
20      "id": 601,
```

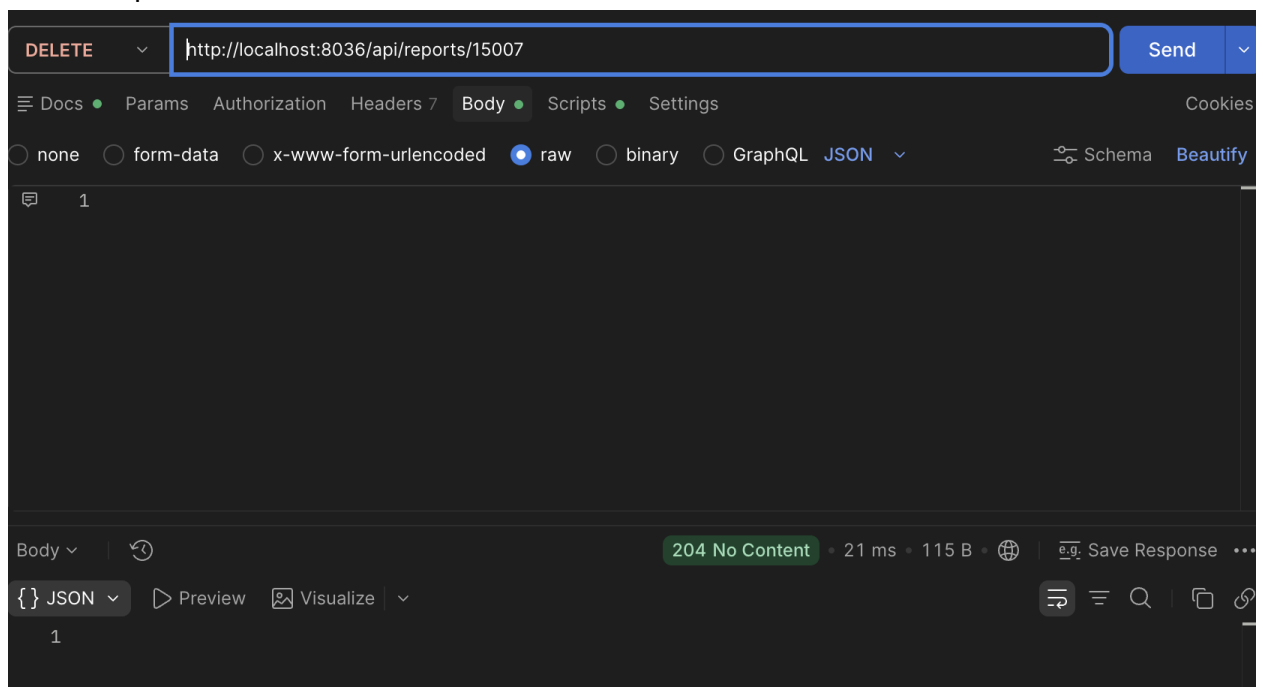
Reports Associated with an Advisory



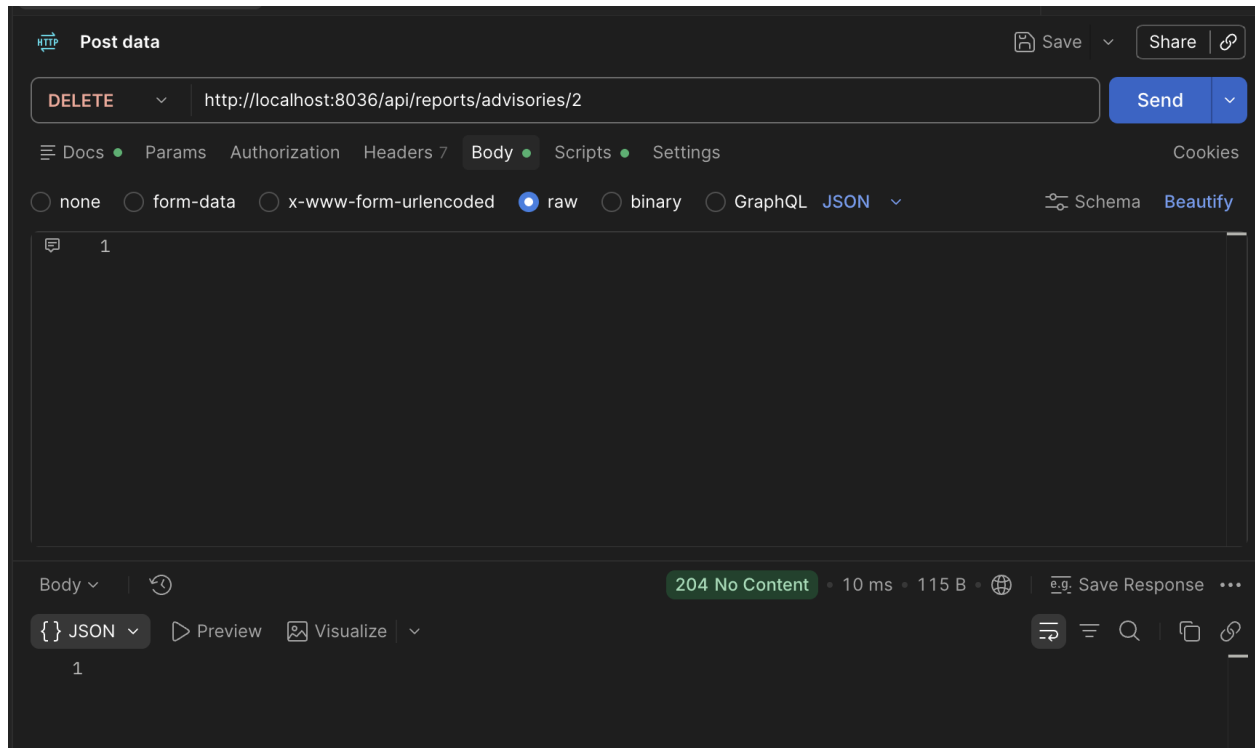
Advisory Deletion blocked while reports are linked



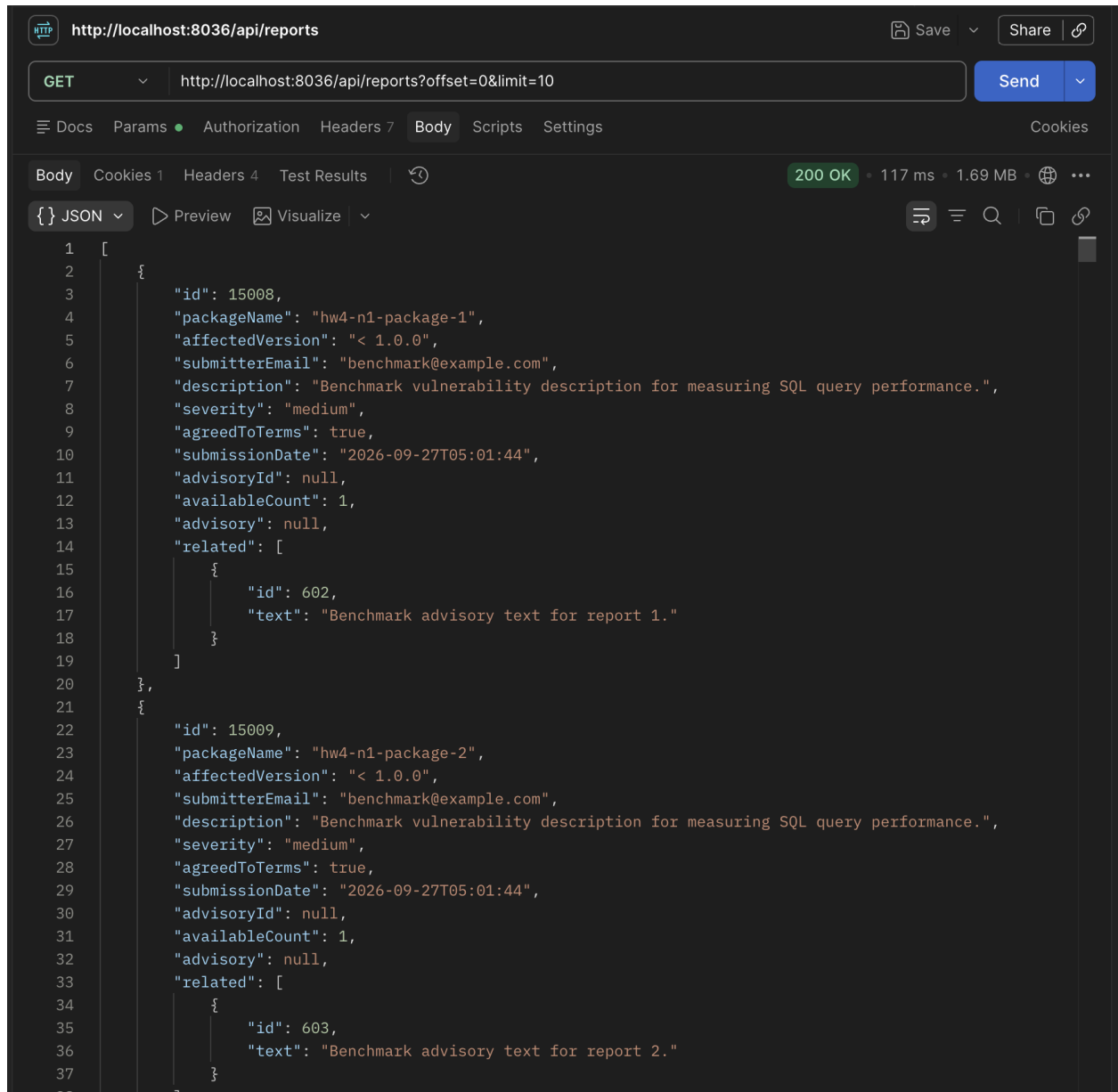
Delete Report API



Delete Advisory API



Report List Pagination



http://localhost:8036/api/reports

GET http://localhost:8036/api/reports?offset=0&limit=10

200 OK • 117 ms • 1.69 MB

```
[
  {
    "id": 15008,
    "packageName": "hw4-n1-package-1",
    "affectedVersion": "< 1.0.0",
    "submitterEmail": "benchmark@example.com",
    "description": "Benchmark vulnerability description for measuring SQL query performance.",
    "severity": "medium",
    "agreedToTerms": true,
    "submissionDate": "2026-09-27T05:01:44",
    "advisoryId": null,
    "availableCount": 1,
    "advisory": null,
    "related": [
      {
        "id": 602,
        "text": "Benchmark advisory text for report 1."
      }
    ]
  },
  {
    "id": 15009,
    "packageName": "hw4-n1-package-2",
    "affectedVersion": "< 1.0.0",
    "submitterEmail": "benchmark@example.com",
    "description": "Benchmark vulnerability description for measuring SQL query performance.",
    "severity": "medium",
    "agreedToTerms": true,
    "submissionDate": "2026-09-27T05:01:44",
    "advisoryId": null,
    "availableCount": 1,
    "advisory": null,
    "related": [
      {
        "id": 603,
        "text": "Benchmark advisory text for report 2."
      }
    ]
  }
]
```

Database Design

```

((.venv) ) ayushishringi@Ayushis-MacBook-Pro data260-0036 % mysql -u root -D s0036_rel -e "SHOW
TABLES; DESCRIBE advisories; DESCRIBE vulnerability_reports;"
+-----+
| Tables_in_s0036_rel |
+-----+
| advisories          |
| related_advisories  |
| sessions            |
| users               |
| vulnerability_reports |
+-----+

+-----+
| Field      | Type          | Null | Key | Default | Extra      |
+-----+
| id         | int           | NO   | PRI | NULL     | auto_increment |
| name       | varchar(255)  | NO   |     | NULL     |               |
| publisher  | varchar(255)  | NO   |     | NULL     |               |
| advisory_code | varchar(80)   | NO   | UNI | NULL     |               |
| created_at | datetime      | NO   |     | NULL     |               |
| updated_at | datetime      | NO   |     | NULL     |               |
+-----+

+-----+
| Field      | Type          | Null | Key | Default | Extra      |
+-----+
| id         | int           | NO   | PRI | NULL     | auto_increment |
| package_name | varchar(255)  | NO   | MUL | NULL     |               |
| affected_version | varchar(255)  | NO   |     | NULL     |               |
| submitter_email | varchar(255)  | NO   |     | NULL     |               |
| description  | text          | NO   |     | NULL     |               |
| severity     | varchar(20)   | NO   |     | NULL     |               |
| agreed_to_terms | tinyint(1)    | NO   |     | NULL     |               |
| submission_date | datetime      | NO   |     | NULL     |               |
| advisory_id  | int           | YES  | MUL | NULL     |               |
| available_count | int           | NO   |     | 1        |               |
+-----+
((.venv) ) ayushishringi@Ayushis-MacBook-Pro data260-0036 %

```

Database Schema Verification

```
((.venv) ) ayushishringi@Ayushis-MacBook-Pro data260-0036 % mysql -u root -D s0036_rel -e "DESCRIBE vulnerability_reports; SHOW INDEX FROM vulnerability_reports; DESCRIBE advisories;"
```

Field	Type	Null	Key	Default	Extra
id	int	NO	PRI	NULL	auto_increment
package_name	varchar(255)	NO	MUL	NULL	
affected_version	varchar(255)	NO		NULL	
submitter_email	varchar(255)	NO		NULL	
description	text	NO		NULL	
severity	varchar(20)	NO		NULL	
agreed_to_terms	tinyint(1)	NO		NULL	
submission_date	datetime	NO		NULL	
advisory_id	int	YES	MUL	NULL	
available_count	int	NO		1	
created_at	datetime	NO		CURRENT_TIMESTAMP	DEFAULT_GENERATED
updated_at	datetime	NO		CURRENT_TIMESTAMP	DEFAULT_GENERATED on update CURRENT_TIMESTAMP
report_code	varchar(80)	NO	UNI	NULL	

Table	Non_unique	Key_name	Seq_in_index	Column_name
Collation	Cardinality	Sub_part	Packed	Null
Index_type	Comment	Index_comment	Visible	Expression
vulnerability_reports	0	PRIMARY	1	id
A	4989	NULL	NULL	BTREE
LL				

Table	Non_unique	Key_name	Seq_in_index	Column_name	Collation	Cardinality	Sub_part	Packed	Null	Index_type	Comment	Index_comment	Visible	Expression
vulnerability_reports	0	PRIMARY	1	id	A	4989			NULL	BTREE			YES	
vulnerability_reports	0	uq_vulnerability_reports_report_code	1	report_code	A	4828			NULL	BTREE			YES	
vulnerability_reports	1	idx_vulnerability_reports_package_name	1	package_name	A	4989			NULL	BTREE			YES	
vulnerability_reports	1	fk_reports_advisory	1	advisory_id	A	1			NULL	BTREE			YES	
Field	Type	Null	Key	Default	Extra									
id	int	NO	PRI	NULL	auto_increment									
name	varchar(255)	NO		NULL										
publisher	varchar(255)	NO		NULL										
advisory_code	varchar(80)	NO	UNI	NULL										
created_at	datetime	NO		NULL										
updated_at	datetime	NO		NULL										

TheMealDB MCP Tool Definitions

```

16 def _get(path: str, params: dict[str, str] | None = None) -> dict[str, Any]:
17     query = "?" + urllib.parse.urlencode(params or {}) if params else ""
18
19     def request() -> dict[str, Any]:
20         with urllib.request.urlopen(BASE + path + query, timeout=5) as response:
21             import json
22             return json.loads(response.read().decode("utf-8"))
23
24     value, _, _ = retry_operation(request, attempts=3, timeout_seconds=5)
25     return value
26
27
28 def _details(meal: dict[str, Any] | None) -> dict[str, Any]:
29     if not meal:
30         return {"message": "no matches"}
31     ingredients = []
32     for index in range(1, 21):
33         name = (meal.get(f"strIngredient{index}") or "").strip()
34         measure = (meal.get(f"strMeasure{index}") or "").strip()
35         if name:
36             ingredients.append({"name": name, "measure": measure})
37     return {"id": meal.get("idMeal"), "name": meal.get("strMeal"), "category": meal.get("strCa"),
38           "area": meal.get("strArea"), "instructions": meal.get("strInstructions"),
39           "image": meal.get("strMealThumb"), "source": meal.get("strSource"),
40           "youtube": meal.get("strYoutube"), "ingredients": ingredients}
41
42
43 def search_meals_by_name(query: str, limit: int = 5) -> list[dict[str, Any]]:
44     if not isinstance(query, str) or not query.strip() or not 1 <= limit <= 25:
45         raise ValueError("query must be non-empty and limit must be 1-25")
46     meals = (_get("/search.php", {"s": query.strip()}).get("meals") or [])[:limit]
47     return [{"id": m.get("idMeal"), "name": m.get("strMeal"), "area": m.get("strArea"),
48           "category": m.get("strCategory"), "thumb": m.get("strMealThumb")} for m in meals]
49
50
51 def meals_by_ingredient(ingredient: str, limit: int = 12) -> list[dict[str, Any]]:
52     if not isinstance(ingredient, str) or not ingredient.strip() or not 1 <= limit <= 25:
53         raise ValueError("ingredient must be non-empty and limit must be 1-25")
54     meals = (_get("/filter.php", {"i": ingredient.strip()}).get("meals") or [])[:limit]
55     return [{"id": m.get("idMeal"), "name": m.get("strMeal"), "thumb": m.get("strMealThumb")}
56           for m in meals]
57
58 def random_meal() -> dict[str, Any]:
59     return _details((_get("/random.php").get("meals") or [None])[0])
60

```

```

43 def search_meals_by_name(query: str, limit: int = 5) -> list[dict[str, Any]]:
44     raise ValueError("query must be non-empty and limit must be 1-25")
45     meals = (_get("/search.php", {"s": query.strip()}).get("meals") or [])[:limit]
46     return [{"id": m.get("idMeal"), "name": m.get("strMeal"), "area": m.get("strArea"),
47             "category": m.get("strCategory"), "thumb": m.get("strMealThumb")} for m in me
48
49
50
51 def meals_by_ingredient(ingredient: str, limit: int = 12) -> list[dict[str, Any]]:
52     if not isinstance(ingredient, str) or not ingredient.strip() or not 1 <= limit <= 25:
53         raise ValueError("ingredient must be non-empty and limit must be 1-25")
54     meals = (_get("/filter.php", {"i": ingredient.strip()}).get("meals") or [])[:limit]
55     return [{"id": m.get("idMeal"), "name": m.get("strMeal"), "thumb": m.get("strMealThumb
56
57
58 def random_meal() -> dict[str, Any]:
59     return _details((_get("/random.php").get("meals") or [None])[0])
60
61
62 def meal_details(id: str) -> dict[str, Any]:
63     return _details((_get("/lookup.php", {"i": str(id)}).get("meals") or [None])[0])
64
65
66 try:
67     from mcp.server.fastmcp import FastMCP
68 except ImportError:
69     FastMCP = None
70
71 if FastMCP:
72     mcp = FastMCP("meals")
73     mcp.tool()(search_meals_by_name)
74     mcp.tool()(meals_by_ingredient)
75     mcp.tool()(random_meal)
76     mcp.tool()(meal_details)
77     if __name__ == "__main__":
78         mcp.run(transport="stdio")
79

```

TheMealDB search_meals_by_name tool

Tools

Search tools...

Paginated

search_meals_by_name

meals_by_ingredient

random_meal

meal_details

×

Results

1- {

2 "id": "52795",

3 "name": "Chicken Handi",

4 "area": "India",

5 "category": "Chicken",

6 "thumb": "https://www.themeadb.com/images/media/meals/wyxwsp1486979827.jpg"

7 }

1- {

2 "id": "53358",

3 "name": "Chicken Mandi",

4 "area": "India",

5 "category": "Chicken",

6 "thumb": "https://www.themeadb.com/images/media/meals/er4d081765186828.jpg"

7 }

1- {

2 "id": "53110",

3 "name": "Sticky Chicken",

4 "area": "Australian",

5 "category": "Chicken",

6 "thumb": "https://www.themeadb.com/images/media/meals/cj56fs1762340001.jpg"

7 }

1- {

2 "id": "52956",

3 "name": "Chicken Congee",

4 "area": "Chinese",

5 "category": "Chicken",

6 "thumb": "https://www.themeadb.com/images/media/meals/15294446352.jpg"

7 }

1- {

2 "id": "52831",

3 "name": "Chicken Karaage",

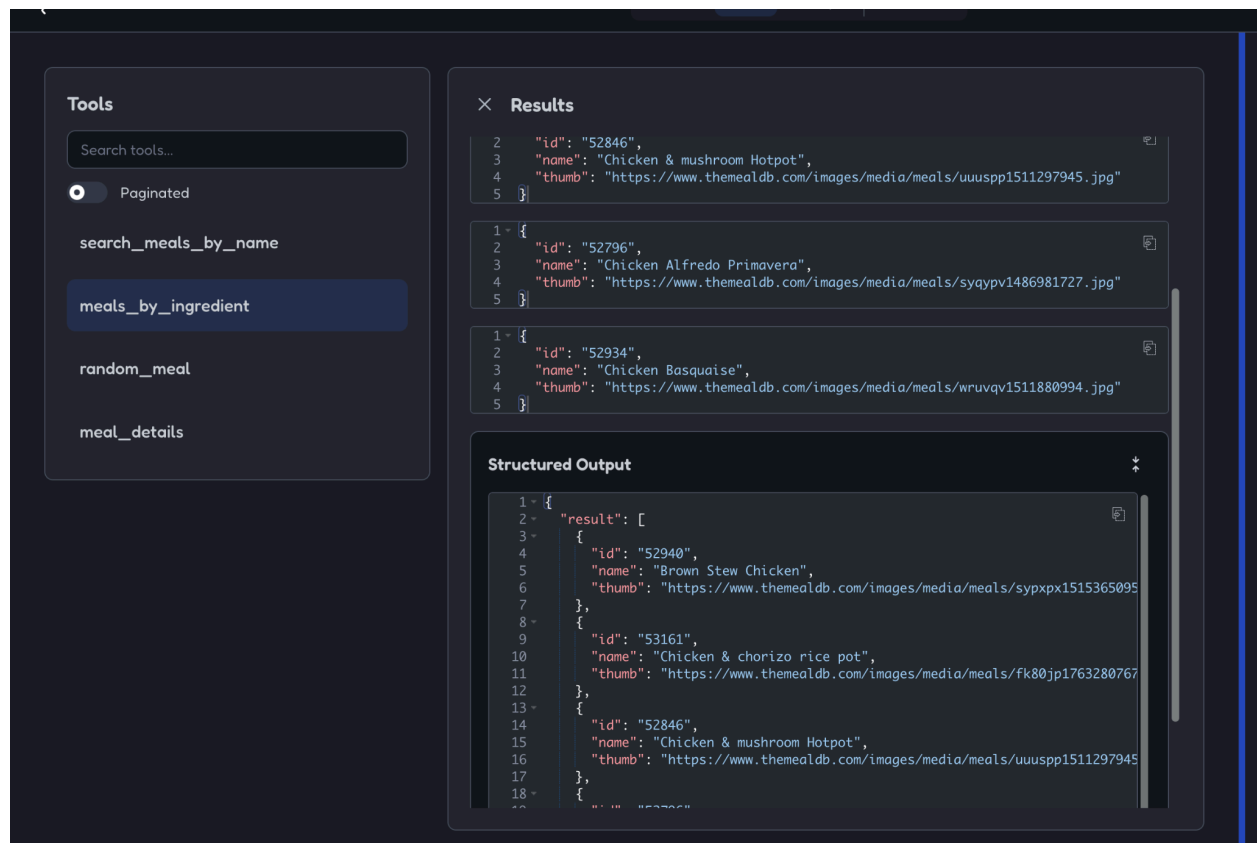
4 "area": "Japanese",

5 "category": "Chicken",

6 "thumb": "https://www.themeadb.com/images/media/meals/tyyws1505930373.jpg"

7 }

TheMealDB meals_by_ingredient tool



TheMealDB random_meal tool

Tools

Search tools...

☐ Paginated

search_meals_by_name

meals_by_ingredient

random_meal

meal_details

×

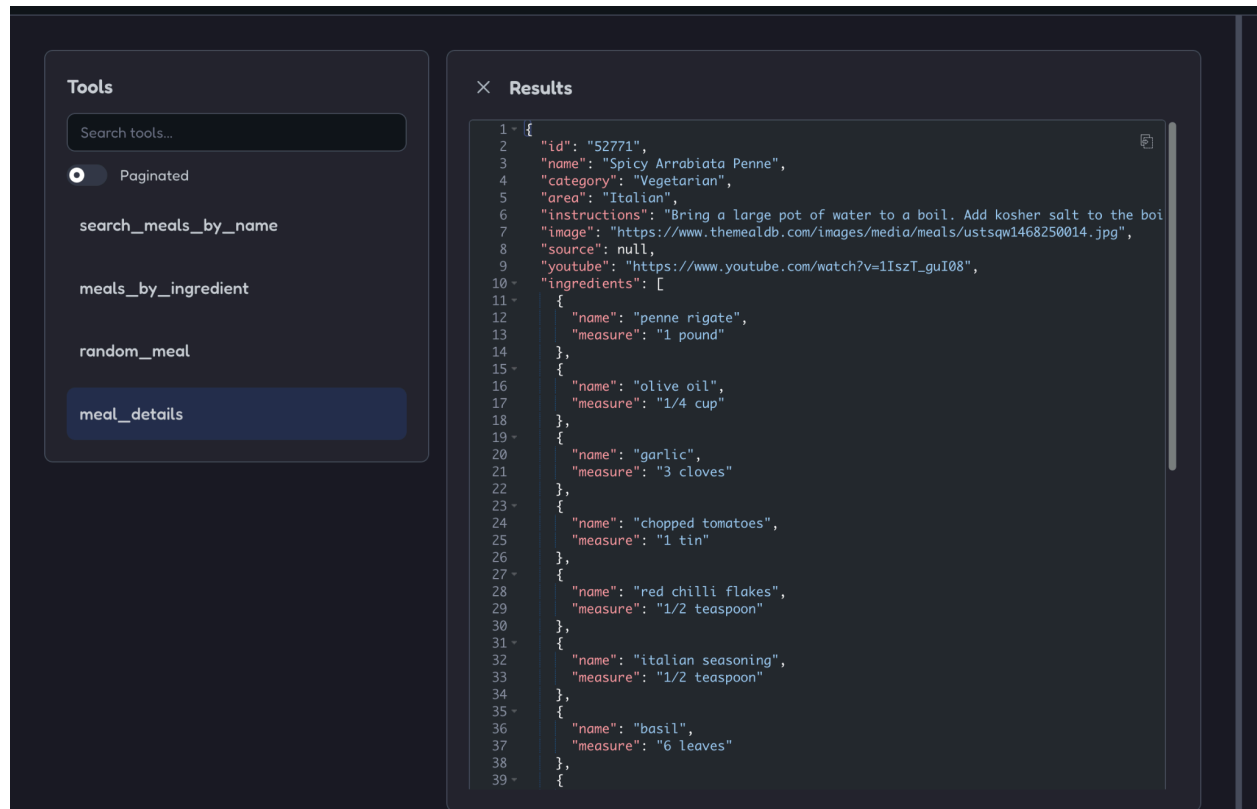
Results

```
1 {
2   "id": "53441",
3   "name": "Banana den Forno",
4   "category": "Dessert",
5   "area": null,
6   "instructions": "Peel the bananas and boil them together with the cinnamon, vanil
7   "image": "https://www.themeadb.com/images/media/meals/6v583k1780093743.jpg",
8   "source": "https://www.visitaruba.com/aruba-recipes/banana-den-forno/",
9   "youtube": "",
10  "ingredients": [
11    {
12      "name": "Banana",
13      "measure": "4"
14    },
15    {
16      "name": "Ground Cinnamon",
17      "measure": "2 tablespoons"
18    },
19    {
20      "name": "Vanilla Extract",
21      "measure": "3  tablespoons"
22    },
23    {
24      "name": "Sugar",
25      "measure": "1 cup"
26    },
27    {
28      "name": "Water",
29      "measure": "4 cups"
30    }
31  ]
32 }
```

Structured Output

```
1 {
2   "id": "53441",
3   "name": "Banana den Forno",
4   "category": "Dessert",
5   "area": null,
6   "instructions": "Peel the bananas and boil them together with the cinnamon, vanil
7   "image": "https://www.themeadb.com/images/media/meals/6v583k1780093743.jpg",
8   "source": "https://www.visitaruba.com/aruba-recipes/banana-den-forno/",
9   "youtube": "",
10  "ingredients": [
11    {
12      "name": "Banana",
13      "measure": "4"
14    },
15    {
16      "name": "Ground Cinnamon",
17      "measure": "2 tablespoons"
18    },
19    {
20      "name": "Vanilla Extract",
21      "measure": "3  tablespoons"
22    },
23    {
24      "name": "Sugar",
25      "measure": "1 cup"
26    },
27    {
28      "name": "Water",
29      "measure": "4 cups"
30    }
31  ]
32 }
```

TheMealDB meal_details tool



Domain MCP Tool Definitions

```

1  # STDIO MCP server for the three s0036_rel domain tools.
2  import logging
3
4  from hw5_tools.domain import DomainStore, envelope
5
6  logging.basicConfig(level=logging.INFO)
7  LOGGER = logging.getLogger("s0036_domain_mcp")
8  store = DomainStore.fixture()
9
10 try:
11     from mcp.server.fastmcp import FastMCP
12 except ImportError: # keeps offline tests/imports usable before pip install mcp[cli]
13     FastMCP = None
14
15 if FastMCP:
16     mcp = FastMCP("s0036_rel")
17
18     @mcp.tool()
19     def search(query: str, limit: int = 5) -> dict:
20         return store.search(query, limit)
21
22     @mcp.tool()
23     def detail(id: str | int) -> dict:
24         return store.detail(id)
25
26     @mcp.tool()
27     def aggregate(severity: str | None = None) -> dict:
28         return store.aggregate(severity)
29
30     if __name__ == "__main__":
31         mcp.run(transport="stdio")
32

```

Domain MCP Search Tool


s0036_rel

Servers
Tools
Prompts
Resources

Tools

Search tools...

☒ Paginated

search

detail

aggregate

Results

```

1 {
2   "ok": true,
3   "data": [
4     {
5       "id": 1,
6       "package": "openssl",
7       "severity": "critical",
8       "available_count": 1,
9       "advisory_code": "CVE-2024-9999"
10    }
11  ],
12  "error": null
13 }

```

Detail MCP Search Tool

s0036_rel

Servers

Tools

Prompts

Resources

Tools

Search tools...

Paginated

search

detail

aggregate

×

Results

1- {

2- "ok": true,

3- "data": {

4- "id": 1,

5- "package": "openssl",

6- "severity": "critical",

7- "available_count": 1,

8- "advisory_code": "CVE-2024-9999"

9- },

10- "error": null

11- }

Domain MCP aggregate tool

s0036_rel

Servers

Tools

Prompts

Resources

Tools

Search tools...

Paginated

search

detail

aggregate

×

Results

1- {

2- "ok": true,

3- "data": {

4- "count": 2,

5- "available_count_total": 4,

6- "by_severity": {

7- "critical": 1,

8- "high": 1,

9- "medium": 0,

10- "low": 0

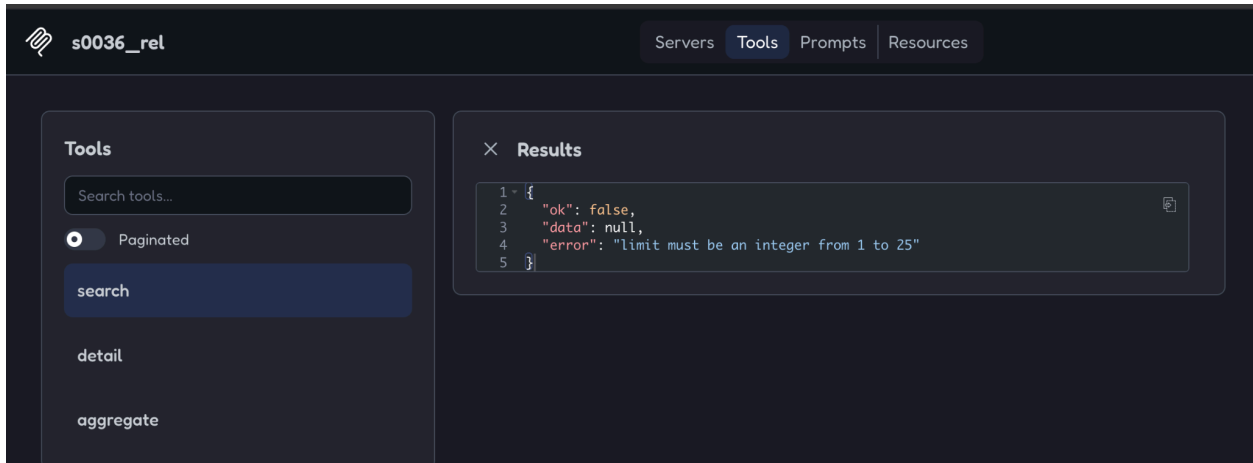
11- }

12- },

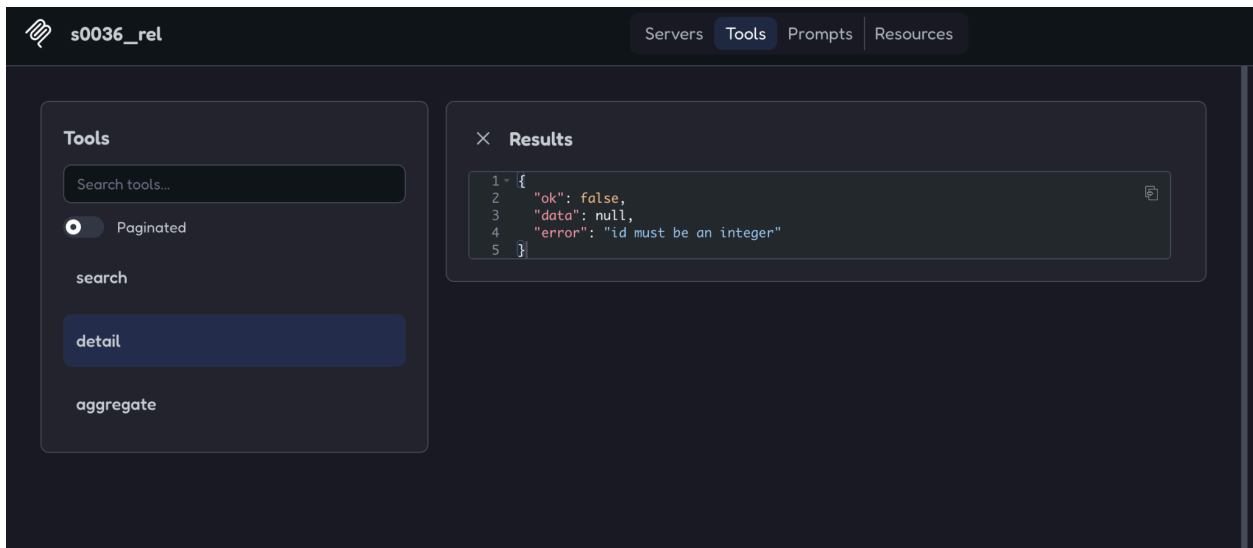
13- "error": null

14- }

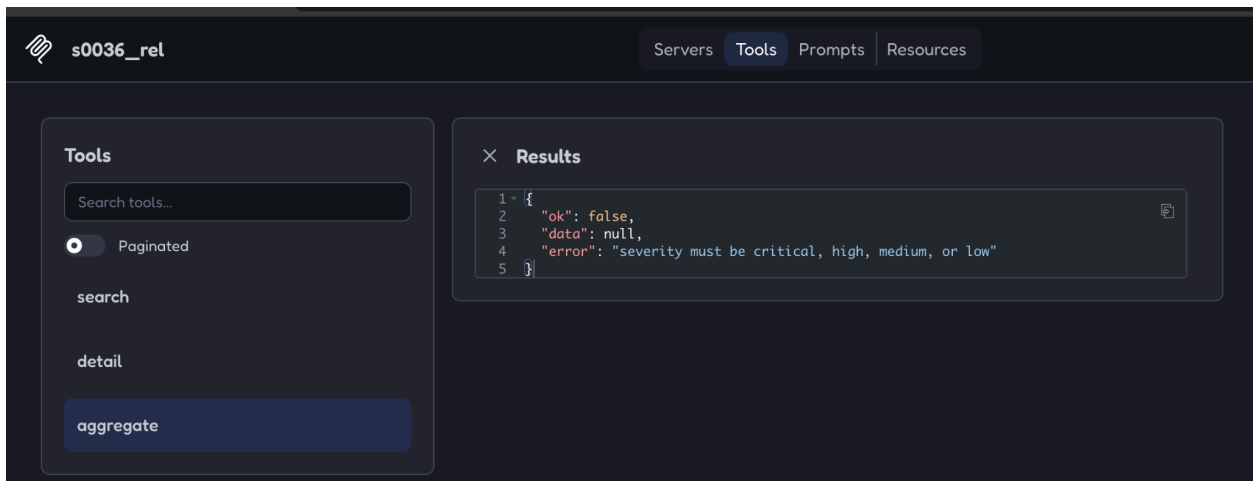
Domain MCP Search Validation Error



Domain MCP Detail Validation Error



Domain MCP Aggregate Validation Error



JSON Validation and Rejected Inputs

Tool	Expected JSON input	Rejected JSON input	Returned error
search	<code>{"query": "openssl"}</code>	<code>{"query": ""}</code>	Copy the exact validation error from your Inspector result
detail	<code>{"id": 1}</code>	<code>{"id": "bad"}</code>	Copy the exact validation error from your Inspector result
aggregate	<code>{"severity": "critical"}</code>	<code>{"severity": "invalid"}</code>	Copy the exact validation error from your Inspector result

Seeded Fault - injection benchmark

```
ault benchmark'
PYTHONPATH=. .venv/bin/python3 scripts/run_hw5_faults.py
2026-10-05 12:11:26 - Final synchronized fault benchmark
[
  {
    "rate": 0.0,
    "success_rate": 1.0,
    "mean_latency_ms": 0.0005,
    "p99_latency_ms": 0.0018
  },
  {
    "rate": 0.2,
    "success_rate": 1.0,
    "mean_latency_ms": 3.436,
    "p99_latency_ms": 37.614
  },
  {
    "rate": 0.5,
    "success_rate": 0.92,
    "mean_latency_ms": 10.9969,
    "p99_latency_ms": 37.5761
  }
]
```

Retry Suitability Analysis

The fault benchmark shows a 100% success rate with no injected failures, 100% success at a 20% failure rate, and a 92% success rate at a 50% failure rate. At the 50% failure rate, mean latency was approximately 10.9969 ms and p99 latency was approximately 37.5761 ms. This is suitable for interactive requests when using a small retry limit, because the latency remains low. For batch processing, I would use approximately 5–8 retries, a 10–30 second timeout, and exponential backoff beginning at 0.5–2 seconds. The benchmark measures simulated fault-injection overhead rather than real external network latency, so production performance may be higher.

Retry and Backoff Implementation

```
83
84 def retry_operation(operation: Callable[[], Any], *, attempts: int = 3,
85                     timeout_seconds: float = 2.0, backoff_seconds: float = 0.01) -> tuple[Any, int, float]:
86     started = time.perf_counter()
87     last_error: Exception | None = None
88     for attempt in range(1, attempts + 1):
89         try:
90             value = operation()
91             return value, attempt, (time.perf_counter() - started) * 1000
92         except Exception as exc:
93             last_error = exc
94             if attempt == attempts:
95                 break
96             time.sleep(min(backoff_seconds * (2 ** (attempt - 1)), timeout_seconds))
97     raise RuntimeError(f"operation failed after {attempts} attempts: {last_error}")
98
99
100 def fault_injection_call(rate: float, call_index: int, seed: int = VERIFY_SEED) -> dict[str, Any]:
101     rng = random.Random(seed + call_index)
102     failures = 0
103
104     def flaky() -> str:
105         nonlocal failures
106         if rng.random() < rate:
107             failures += 1
108             raise OSError("injected transient failure")
109         return "ok"
110
111     started = time.perf_counter()
112     try:
113         value, attempts, _ = retry_operation(flaky)
114         status, error = "success", None
115     except RuntimeError as exc:
116         value, attempts, error = None, 3, str(exc)
117         status = "failure"
118     return {"rate": rate, "call": call_index, "status": status, "attempts": attempts,
119           "injected_failures": failures, "latency_ms": round((time.perf_counter() - started) * 1000, 4),
120           "result": value, "error": error}
121
```

Retry Demonstration

```

• ((.venv) ) ayushishringi@Ayushis-MacBook-Pro data260-0036 % PYTHONPATH=. .venv/bin/python3 scrip
ts/run_retry_cases.py
CASE 1: first attempt succeeds
status=success attempts=1
CASE 2: first attempt fails, retry succeeds
status=success attempts=2
CASE 3: all retries fail
status=failure attempts=3
error=operation failed after 3 attempts: injected persistent failure
((.venv) ) ayushishringi@Ayushis-MacBook-Pro data260-0036 %

```

Offline execute_tool test runner

```

• ((.venv) ) ayushishringi@Ayushis-MacBook-Pro data260-0036 % PYTHONPATH=. .venv/bin/python3 s
cripts/run_hw5_tests.py
PASS search valid
PASS search invalid query
PASS detail valid
PASS detail invalid id
PASS aggregate valid
PASS aggregate invalid severity
PASS safety rule blocked
PASS agent max_steps
8/8 tests passed
((.venv) ) ayushishringi@Ayushis-MacBook-Pro data260-0036 %

```

Safety Rule Enforcement

```

try:
    # Safety rule: aggregate calls cannot request a negative inventory view.
    if name == "aggregate" and inputs.get("include_unavailable") is True:
        return json.dumps(envelope(False, error="safety rule: unavailable inventory is restri
handlers: dict[str, Callable[..., dict[str, Any]]] = {
    "search": store.search,
    "detail": store.detail,
    "aggregate": store.aggregate,
}
if name not in handlers:
    return json.dumps(envelope(False, error=f"unknown tool: {name}"))
return json.dumps(handlers[name](**inputs), sort_keys=True)
except TypeError as exc:
    return json.dumps(envelope(False, error=f"invalid inputs: {exc}"), sort_keys=True)
except Exception as exc: # tool boundary must never crash its caller
    LOGGER.exception("tool failure")
    return json.dumps(envelope(False, error=f"tool failure: {exc}"), sort_keys=True)

```

Allowed and safety-blocked tool calls

```

• ((.venv) ) ayushishringi@Ayushis-MacBook-Pro data260-0036 % PYTHONPATH=. .venv/bin/python -c 'fr
om hw5_tools.domain import execute_tool; print("ALLOWED:", execute_tool("search", {"query": "ope
nssl", "limit": 1})); print("BLOCKED:", execute_tool("aggregate", {"include_unavailable": True})
)'
ALLOWED: {"data": [{"advisory_code": "CVE-2024-9999", "available_count": 1, "id": 1, "package":
"openssl", "severity": "critical"}], "error": null, "ok": true}
BLOCKED: {"ok": false, "data": null, "error": "safety rule: unavailable inventory is restricted"
}
((.venv) ) ayushishringi@Ayushis-MacBook-Pro data260-0036 %

```

Safe Tool Execution Boundary

```
56         "by_severity": {level: sum(r["severity"] == level for r in rows)
57                           for level in ("critical", "high", "medium", "low")},
58     })
59
60
61 def execute_tool(name: str, inputs: dict[str, Any] | None, store: DomainStore | None = None) -> str:
62     """The only entry point exposed to the Part 5 agent."""
63     store = store or DomainStore.fixture()
64     inputs = inputs or {}
65     try:
66         # Safety rule: aggregate calls cannot request a negative inventory view.
67         if name == "aggregate" and inputs.get("include_unavailable") is True:
68             return json.dumps(envelope(False, error="safety rule: unavailable inventory is restricted"))
69         handlers: dict[str, Callable[..., dict[str, Any]]] = {
70             "search": store.search,
71             "detail": store.detail,
72             "aggregate": store.aggregate,
73         }
74         if name not in handlers:
75             return json.dumps(envelope(False, error=f"unknown tool: {name}"))
76         return json.dumps(handlers[name](**inputs), sort_keys=True)
77     except TypeError as exc:
78         return json.dumps(envelope(False, error=f"invalid inputs: {exc}"), sort_keys=True)
79     except Exception as exc: # tool boundary must never crash its caller
80         LOGGER.exception("tool failure")
81         return json.dumps(envelope(False, error=f"tool failure: {exc}"), sort_keys=True)
82
83
84 def retry_operation(operation: Callable[[], Any], *, attempts: int = 3,
85                    timeout_seconds: float = 2.0, backoff_seconds: float = 0.01) -> tuple[Any, int, flo
```

Bounded Agent Loop

```

23     return call
24
25
26 def run_agent(user_input: str, model: Any | None = None, *, max_steps: int = 5,
27              store: DomainStore | None = None, log_path: str | Path = "agent_runs.jsonl") -> dict[str,
28              model = model or MockModel()
29              store = store or DomainStore.fixture()
30              history: list[dict[str, Any]] = []
31              stop_reason = "max_steps"
32              for step in range(1, max_steps + 1):
33                  call = model.next_call(user_input, history)
34                  if call is None:
35                      stop_reason = "normal_completion"
36                      break
37                  result = json.loads(execute_tool(call.get("name", ""), call.get("inputs", {}), store))
38                  event = {"step": step, "user_input": user_input, "tool": call.get("name"),
39                          "inputs": call.get("inputs", {}), "result": result}
40                  history.append(event)
41                  with Path(log_path).open("a", encoding="utf-8") as stream:
42                      stream.write(json.dumps(event, sort_keys=True) + "\n")
43                  if not result["ok"] and result["error"].startswith("safety rule"):
44                      st (parameter) log_path: str | Path
45                      br
46                  summary = log_path: str | Path = "agent_runs.jsonl" (history), "stop_reason": stop_reason}
47                  with Path(log_path).open("a", encoding="utf-8") as stream:
48                      stream.write(json.dumps({"final": summary}, sort_keys=True) + "\n")
49                  return {**summary, "events": history}
50

```

Agent-loop scenario results

Scenario	Steps	Tool calls	Stop reason
Normal completion	1	1	normal_completion
Maximum-step limit	2	2	max_steps
Safety-rule block	1	1	safety_rule_block
Invalid tool input	1	1	normal_completion

```

((.venv) ) ayushishringi@Ayushis-MacBook-Pro data260-0036 % cat reports/hw05/raw/agent_scenarios
.json
[
  {
    "scenario": "normal_completion",
    "steps": 1,
    "stop_reason": "normal_completion",
    "tool_call_count": 1
  },
  {
    "scenario": "max_steps",
    "steps": 2,
    "stop_reason": "max_steps",
    "tool_call_count": 2
  },
  {
    "scenario": "safety_rule_block",
    "steps": 1,
    "stop_reason": "safety_rule_block",
    "tool_call_count": 1
  },
  {
    "scenario": "invalid_tool_input",
    "steps": 1,
    "stop_reason": "normal_completion",
    "tool_call_count": 1
  }
]
((.venv) ) ayushishringi@Ayushis-MacBook-Pro data260-0036 %

```

These four recorded scenarios are from the offline harness and use MockModel. The configured live model is qwen3 :4bthrough Ollama; therefore, the offline scenario metrics should not be presented as live Ollama measurements.

Fault Injection Metrics

Failure rate	Success rate	Mean latency (ms)	p99 latency (ms)
0%	100%	0.0005	0.0018
20%	100%	3.4360	37.6140
50%	92%	10.9969	37.5761

As the injected failure rate increased, latency increased and success rate decreased. The retry policy maintained 100% success at a 20% failure rate, while the 50% failure rate resulted in 92% success.

Retry behavior:

A successful call returns on the first attempt when no transient failure is injected. If the first attempt fails, `retry_operation` retries the call with bounded exponential backoff. If all three attempts fail, the function returns a clean `RuntimeError` instead of crashing the caller.

Tool input and validation:

The search tool expects `{"query": string, "limit": integer}`. The detail tool expects `{"id": integer or string containing an integer}`. The aggregate tool expects an optional `{"severity": "critical" | "high"}`

| "medium" | "low"} object. Invalid inputs, such as an empty query, a limit outside 1–25, a non-integer detail ID, or severity "urgent", return the standard error envelope {"ok": false, "data": null, "error": "..."}.

Policy comparison:

Interactive applications should use a small number of retries and short capped backoff so users receive a quick response. Batch processing can use more retries and a longer capped backoff because reliability is more important than immediate response time.

Reflection

In the selected agent run, the model proposed a bounded search call twice. The harness passed each call through `execute_tool`, which returned the shared {ok, data, error} envelope. Each tool call was recorded with its step number, inputs, tool name, and result. Because `max_steps` was set to 2, the agent stopped after two calls with `stop_reason`: "max_steps" and `tool_call_count`: 2. No exception escaped the tool boundary. This run demonstrates why a step limit is necessary: it prevents an agent from continuing indefinitely when the model keeps proposing additional tool calls. The recorded JSONL event and final summary make the stopping behavior auditable.

AI USE

```
# AI use disclosure

1. I used an AI assistant to inspect the HW4 repository, scaffold the HW5
API/MCP/tool-test structure, and review integration points. I selected the domain
model, checked the assignment requirements, ran the verification commands, and remain
responsible for the implementation and evidence.

2. I compared two seeded fault-injection runs using VERIFY_SEED=260036. The success
rates remained consistent: 100% at 0% and 20% injected failure, and 92% at 50%
failure. The latency values varied slightly between runs, so I used the final
synchronized run for the report, raw JSON/CSV files, metrics table, and screenshot. I
verified the outputs manually instead of accepting AI-generated values without
checking them.

3. I detected this by running the same seeded fault benchmark twice, comparing the
machine-readable sequences, and checking that server logging is configured through
Python logging rather than `print`.

4. I kept the seed explicit in every benchmark call and routed diagnostics through
logging. The offline test suite also exercises invalid inputs, the safety rule, and
the max-step ceiling without a model or network.
```

Run Log

HW5 Run Log

HW5 run log

Configuration: SID4=0036 PORT_BASE=8036 PREFIX=s0036 SEED=36

VERIFY_SEED=260036

Run commands:

PYTHONPATH=. python3 scripts/run_hw5_tests.py

PYTHONPATH=. python3 scripts/run_hw5_faults.py

PYTHONPATH=. python3 scripts/verify_hw5.py

The commands produce the raw fault-injection CSV/JSON and verification.json in this directory.

Final synchronized benchmark run:

2026-10-05 09:28:33 - Running fault benchmark

PYTHONPATH=. .venv/bin/python3 scripts/run_hw5_faults.py

Failure rate	Success rate	Mean latency (ms)	p99 latency (ms)
0%	100%	0.0005	0.0018
20%	100%	3.4360	37.6140
50%	92%	10.9969	37.5761

2026-10-04 16:09:38 - Running final verification

passed: true

all_required_files_present: true

commit_hash:02672d93befa5c559f93698fd8d3a1881461d7f1

Final Verification

```

((.venv) ) ayushishringi@Ayushis-MacBook-Pro data260-0036 % make verify-hw05
PYTHONPATH=. .venv/bin/python3 scripts/verify_hw5.py
{
  "passed": true,
  "homework": "DATA-260 HW5",
  "SID4": "0036",
  "PORT_BASE": 8036,
  "PREFIX": "s0036",
  "SEED": 36,
  "VERIFY_SEED": 260036,
  "DOMAIN_ID": 4,
  "model": "qwen3:4b (Ollama; offline tests use MockModel)",
  "checks": {
    "offline_tool_tests": {
      "passed": true,
      "output": "PASS search valid\nPASS search invalid query\nPASS detail valid\nPASS detail invalid id\nPASS aggregate valid\nPASS aggregate invalid severity\nPASS safety rule blocked\nPASS agent max_steps\n8/8 tests passed\n"
    },
    "required_files": {
      "backend/models.py": true,
      "backend/report_api.py": true,
      "frontend/src/store.js": true,
      "domain_mcp_server.py": true,
      "meals_server.py": true,
      "scripts/run_hw5_tests.py": true,
      "scripts/run_hw5_faults.py": true,
      "scripts/run_retry_cases.py": true,
      "reports/hw05/METRICS.md": true
    },
    "all_required_files_present": true,
    "runtime_entrypoints": {
      "fastapi_app_import": true,
      "fastapi_health_route": true,
      "fastapi_reports_route": true,
      "domain_mcp_import": true,
      "domain_mcp_tools": [
        "aggregate",
        "detail",
        "search"
      ],
      "domain_mcp_tools_complete": true,
      "meals_mcp_import": true,
      "meals_mcp_tools": [
        "meal_details",
        "meals_by_ingredient",
        "random_meal",
        "search_meals_by_name"
      ],
      "meals_mcp_tools_complete": true,
      "runtime_entrypoints_ready": true
    }
  }
}
}
((.venv) ) ayushishringi@Ayushis-MacBook-Pro data260-0036 %

```